

Notes from the meeting with Lux:

Web ocean (cloud servers) has their servers.

Refreshing every half hour (what made you decide on that? *Better delayed*. I am not sure I agree with that)

Why the weight 1 for a month, decrease over 5 months down to a weight of a half? *No solid reasoning, this is flexible*. I would recommend exponentially decreasing weights with time for algorithmic simplicity.

Programming languages:

- React with typescript for front end (it is JavaScript library)

- Sql database (SQ lite), VS Code

Investor is called Neil

Note, I want to look into gaming leaderboards to see how they do their averaging (Editorial note: I didn't do this research.)

Range for numbers from 0-100 based on four category sliders that users will interact with.

Four categories for rating by users for bands (acts/performances/etc.): performance, musicianship, audience engagement, originality (0-25 on each)

Four categories for venues: sound, ambience, staff, amenities

Let $\bar{x}_n$ be the running weighted average through the n^{th} data point:

$$\bar{x}_n = \frac{N_n}{D_n} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

Here N_n is the numerator of the running average through the n^{th} data point, D_n is the denominator of the running average through the n^{th} data point, and w_i is the weight of the i^{th} data point.

Then $\bar{x}_{n+1}$ has the following form:

$$\bar{x}_{n+1} = \frac{N_{n+1}}{D_{n+1}} = \frac{\sum_{i=1}^{n+1} w_i x_i}{\sum_{i=1}^{n+1} w_i} = \frac{w_{n+1} x_{n+1} + \sum_{i=1}^n w_i x_i}{w_{n+1} + \sum_{i=1}^n w_i} = \frac{w_{n+1} x_{n+1} + N_n}{w_{n+1} + D_n}$$

This formulation makes an assumption however that will not work for you. It assumes that the weights do not change with time. I recommend an exponentially decreasing weight scheme to accomplish this since we want the older data points to be worth less than the newest ones to allow for bands (venues) to not be overly burdened (or boosted) by their past performances. In this way, the current weighted average gives a more accurate current rating of each band. There are many choices for a decreasing weighting scheme, but exponential decay has certain niceties because of the properties of the exponential as I'll show now.

Let $w_i = \left(\frac{1}{2}\right)^{\frac{a_i}{T}}$ where a_i is the age of the data and T is the period of time that represents the “half-life” of the data point weight since when the age $a_i = 0$, $w_i = 1$, and when $a_i = T$, $w_i = \frac{1}{2}$. Another nicety of this style of weighting is that there will be no change in the average if all the ages increase by the same amount Δt . Notice, due to :

$$\begin{aligned} D_n &= \sum_{i=1}^n w_i = \sum_{i=1}^n \left(\frac{1}{2}\right)^{\frac{a_i}{T}} \xrightarrow{a_i \rightarrow a_i + \Delta t} \sum_{i=1}^n \left(\frac{1}{2}\right)^{\frac{a_i + \Delta t}{T}} = \sum_{i=1}^n \left(\frac{1}{2}\right)^{\frac{a_i}{T}} \left(\frac{1}{2}\right)^{\frac{\Delta t}{T}} = \left(\frac{1}{2}\right)^{\frac{\Delta t}{T}} \sum_{i=1}^n \left(\frac{1}{2}\right)^{\frac{a_i}{T}} \\ &= \left(\frac{1}{2}\right)^{\frac{\Delta t}{T}} D_n \end{aligned}$$

So, the denominator accrues a factor of $\left(\frac{1}{2}\right)^{\frac{\Delta t}{T}}$. By the same logic, the numerator N_n also accrues this factor, so these factors cancel out in the average.

Now, let's think about updating the average when a new datapoint x_{n+1} is added with timestamp t_{n+1} . To start, we need a way to relate the age of a datapoint to its timestamp. Giving the newly added datapoint an age of 0 will give it full weight, and earlier timestamps should have greater age; hence we'll adopt $a_{i,n+1} = t_{n+1} - t_i$. This age will need to be

adjusted whenever a new datapoint is added, but because of the above exponent properties, it turns out to be a rather simple adjustment.

Assume that we have already calculated a current weighted average $\bar{x}_n = \frac{N_n}{D_n}$. The denominator will take the form

$$D_n = \sum_{i=1}^n w_i = \sum_{i=1}^n \left(\frac{1}{2}\right)^{\frac{a_{i,n}}{T}} = \sum_{i=1}^n \left(\frac{1}{2}\right)^{\frac{t_n - t_i}{T}}$$

The new denominator will take the form

$$\begin{aligned} D_{n+1} &= \sum_{i=1}^{n+1} w_i = \sum_{i=1}^{n+1} \left(\frac{1}{2}\right)^{\frac{a_{i,n+1}}{T}} = \sum_{i=1}^{n+1} \left(\frac{1}{2}\right)^{\frac{t_{n+1} - t_i}{T}} \\ &= \sum_{i=1}^{n+1} \left(\frac{1}{2}\right)^{\frac{t_{n+1} - t_n + t_n - t_i}{T}} \\ &= \left(\frac{1}{2}\right)^{\frac{t_{n+1} - t_n + t_n - t_{n+1}}{T}} + \sum_{i=1}^n \left(\frac{1}{2}\right)^{\frac{t_{n+1} - t_n + t_n - t_i}{T}} \\ &= \left(\frac{1}{2}\right)^0 + \sum_{i=1}^n \left(\frac{1}{2}\right)^{\frac{t_{n+1} - t_n}{T}} \left(\frac{1}{2}\right)^{\frac{t_n - t_i}{T}} \\ &= 1 + \left(\frac{1}{2}\right)^{\frac{t_{n+1} - t_n}{T}} \sum_{i=1}^n \left(\frac{1}{2}\right)^{\frac{t_n - t_i}{T}} \\ &= 1 + \left(\frac{1}{2}\right)^{\frac{\Delta t}{T}} D_n \end{aligned}$$

Where $\Delta t = t_{n+1} - t_n$. Similarly, the new numerator will take the form

$$N_{n+1} = x_{n+1} + \left(\frac{1}{2}\right)^{\frac{\Delta t}{T}} N_n$$

So the new average will be

$$\bar{x}_{n+1} = \frac{N_{n+1}}{D_{n+1}} = \frac{x_{n+1} + \left(\frac{1}{2}\right)^{\frac{\Delta t}{T}} N_n}{1 + \left(\frac{1}{2}\right)^{\frac{\Delta t}{T}} D_n}$$

With this method, each time you calculate the new average, you will have to have to stored t_n (in order to calculate Δt), N_n , and D_n , and you will need to use the new datapoint x_{n+1}

and timestamp t_{n+1} . However, since you are already going to be storing $\bar{x}_n$, so can rearrange in the following way to save on some compute.

$$\bar{x}_{n+1} = \frac{N_{n+1}}{D_{n+1}} = \frac{x_{n+1} + \left(\frac{1}{2}\right)^{\frac{\Delta t}{T}} N_n}{1 + \left(\frac{1}{2}\right)^{\frac{\Delta t}{T}} D_n} = \frac{x_{n+1} + \left(\frac{1}{2}\right)^{\frac{\Delta t}{T}} D_n \frac{N_n}{D_n}}{1 + \left(\frac{1}{2}\right)^{\frac{\Delta t}{T}} D_n} = \frac{x_{n+1} + \left(\frac{1}{2}\right)^{\frac{\Delta t}{T}} D_n \bar{x}_n}{1 + \left(\frac{1}{2}\right)^{\frac{\Delta t}{T}} D_n}$$

This rearrangement gets rid of the need to store the numerator and allows for the following kind of updating function for the average (assuming `x_ave`, `t_current`, and `denominator` are global to this function):

```
updateAverage(double x_next, double t_next) {
    double decayed_denominator = (1/2)**((t_next -
t_current)/half_life)*denominator;

    denominator = 1+decayed_denominator;

    x_ave = (x_next+decayed_denominator*x_ave)/(1+denominator);

    t_current = t_next;
}
```

A couple of things to note. This kind of function doesn't use that much compute so it is likely that you will be able to update things real time. However, it is good to design a buffer so that you can still compute it one at a time when you get a lot of new ratings all at once.

There are also ways to update with multiple ratings in the buffer. I can work that out if you would like, but it doesn't save on that much compute. However, it would allow for bulk computing less frequently if that is the preference. Let me know. I think it makes more sense for me that these averages per band/venue happen in real time. Leaderboards are another thing entirely, and those would make sense to me both in real time or more spaced out depending on your software infrastructure.

Another thing to note, you should use a floating point variable which contains enough precision to handle these exponential calculations and deal with the denominator becoming a larger number. Below I've used "double" floating point variables in my pseudocode, but other choices may be better depending on how things are structured in your code. This method will likely accrue some floating point error which will change it from the true value especially if you just let it run for a long time without recalibrating. The nice thing about this kind of method though is that you can always do some recalibrating using raw data:

$$\bar{x}_n = \frac{N_n}{D_n} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

You will need to choose T so that the decrease in weight makes sense for the band and venues. It seems quite possible that different T -values should be considered for bands and venues since bands are likely to be more volatile in their ratings than venues will be. One nice thing about exponentially decreasing weights is that the numerators and denominators of the average calculation take a while to get out of hand. Additionally, choosing a smaller half-life will also allow for quicker decay of rating weights which can help that as well. You don't want to go too small though. I would perhaps recommend $T =$ approximately a year so that 1-year-old rating has half the weight of present ratings, and 2-year-old ratings have weight of $\frac{1}{4}$, and 3-year-old rating have weight of $\frac{1}{8}$ and so on with decreasing fractional powers of 2.

Finally, it is totally possible I've made some mistake in here, if you find one or have any other questions for me, feel free to reach out.

-Brandon James Johnson